

Evaluation Problem 2

Group C21

Duration: 2 hrs

Instruction:

Imagine an office with a single bathroom. The following rules must be enforced by your program:

Mutual Exclusion of Groups:

- If a Woman is in the bathroom, any other Woman can enter (up to the capacity).
- If a Woman is in the bathroom, no Man can enter. He must wait outside.
- Conversely, if a Man is inside, no Woman can enter.

Capacity Limit (N):

The bathroom only has N stalls. Even if only one gender is using it, the total number of people inside cannot exceed N .

Starvation Prevention (Fairness):

If the bathroom is full of Women and there is a constant stream of new Women arriving, a Man waiting outside might never get a chance to enter. This is called Starvation.

The Rule: If a member of the opposite gender is waiting, you must limit how many people of the current gender can enter consecutively (let's call this limit X).